AGS SA-NT
Chapter Seminar 2021

AUSTRALIAN GEOMECHANICS SOCIETY

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Static Liquefaction of Soils and Tailings

SPEAKER:
Dr Mizanur Rahman, Associate Professor in Geotechnical Engineering, University of South Australia

ABSTRACT

Seismically induced liquefaction failures of natural soils and mine tailings has been extensively covered in the literature. However, static liquefaction which occurs in the absence of seismic or dynamic loading is less known to the engineers but often led to devastating consequences including multiple fatalities, e.g. Samarco dam failure in Brazil (2015), Merriespruit failure in South Africa (1994). Static liquefaction may be triggered by many different static loading conditions, such as the rapid raising of perimeter embankments, reduction of mean effective stress due to a rising phreatic surface, or over steepening of the embankment slope, among many others. In this presentation, I will show simple elemental soil tests, such as triaxial and direct simple shear, for understanding the triggering mechanism of static liquefaction, to predict static liquefaction susceptibility and their remedial measure under the critical state soil mechanics framework. The use of field tests. e.g. SPT and CPT, to identify liquefaction susceptibility will be presented. The unknown and the limitation in the current state of knowledge and associate risk will be discussed.

SPEAKER BIOGRAPHY

Dr Md Mizanur Rahman is an Associate Professor in Geotechnical Engineering and he currently leads a research group of 3 academics, 1 research associate and 15 PhD students at the University of South Australia. He has a strong track record of fundamental research into many aspects of geotechnical and pavement engineering with the support of prestigious fellowships and federal/state government funding. These include an ARC Linkage grant (LP160101561) for a study on 'Evaluating potential static liquefaction of tailings to prevent failures' which is a continuation of his PhD dissertation on 'Modelling the effect of fines on liquefaction behaviour of sandy soil' from the University of New South Wales. His current research interest includes liquefaction assessment, pavement engineering, recycling materials for the circular economy, soil reactivity and failure of buried pipelines. The results of these research studies have been published in many reports and more than 160 peer-reviewed articles. He has maintained high-level professional engagement in both academia and practice via being editor-in-chief and editorial board member of several international academic journals. He is also a current member of three technical committees of the International Society for Soil Mechanics and Geotechnical Engineering. As part of his pavement research, he recently authored the DesignPave software, which is now the national standard software for the design of Interlocking and Permeable Paving and is freely distributed through the Concrete Masonry Association of Australia (CMAA).

WHEN
Thursday 29 April 2021

TIME
5:30PM – 7:00PM

Light refreshments will be served at 5:30pm for a 6pm presentation start

WHERE:
Room N132, Engineering North, University of Adelaide, North Terrace Campus

COST:
Free

EVENT CONTACT:
For further information please contact Xiyang Chen via xiyang.chen@adelaide.edu.au

RSVP:
Please RSVP via: https://australiangeomechanics.org/meetings/static-liquefaction-of-soils-and-tailings/

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